

DAY 02 - I: DEEP DIVE INTO OUTLIERS

Bilingual Study Notes (English & Hindi) | Mission 2029

1. What is an Outlier? (Outlier Kya Hai?)

An outlier is a data point that is significantly different from the remaining data points in a dataset.

Outlier ek aisi value hoti hai jo baki saare data points se bahut zyada alag (bahut badi ya bahut choti) hoti hai.

It is like a "Stranger" in a group that doesn't fit the normal pattern.

Ye ek dher mein us "ajnabi" ki tarah hai jo normal pattern se mel nahi khata.

2. Why do Outliers occur? (Outliers Kyun Hote Hain?)

1. Measurement Error: Faulty equipment or manual entry mistake.

1. Measurement Error: Machine ki kharabi ya likhne mein galti (manav galti).

2. Natural Variation: Rare but real events (e.g., an exceptionally heavy gold order).

2. Natural Variation: Kabhi-kabhi hone wali asli ghatnayein (jaise sone ka koi bahut bada order).

3. Effect on Mean (Mean Par Asar)

Outliers "pull" the mean towards themselves, giving a false picture of the average.

Outliers mean ko apni taraf "khinch" lete hain, jisse average ki galat tasveer milti hai.

Scenario	Data (Ring Weights)	Calculated Mean
Normal Data	4g, 5g, 6g, 5g, 5g	5g (Sahi representative)
With Outlier	4g, 5g, 6g, 5g, 80g	20g (Galat representative)

Yahan 80g ek outlier hai jisne average ko 5g se badha kar 20g kar diya.

4. How to handle Outliers? (Outliers Se Kaise Niptien?)

1. Investigate: Check if it's a typing error. If yes, correct it.

1. Investigate: Check karein ki ye typing ki galti toh nahi. Agar hai toh sahi karein.

2. Remove: If it's a mistake, delete the outlier.

2. Remove: Agar wo galti hai, toh us outlier ko data se hata dein.

3. Use Median: When outliers exist, use 'Median' instead of 'Mean' for better accuracy.

3. Use Median: Jab data mein outliers hon, toh 'Mean' ki jagah 'Median' ka istemal karein kyunki wo zyada bhorosemand hota hai.

Origin Tip: As an Analyst, never trust the Mean blindly. Always check for Outliers first!

Origin Tip: Ek Analyst ke taur par, kabhi bhi Mean par andha bharosa na karein. Pehle hamesha Outliers check karein!